

Module 57.1: Yield-Based Bond Duration Measures and Properties

SCHWESERNOTES - BOOK 3

LOS 57.a: Define, calculate, and interpret modified duration, money duration, and the price value of a basis point (PVBP).

Modified duration (ModDur) is calculated as Macaulay duration (MacDur) divided by one plus the bond's periodic rate of return YTM divided by periodicity).

For an annual-pay bond, this is the general form of ModDur:

$$\text{ModDur} = \text{MacDur} / (1 + \text{YTM})$$

For a semiannual-pay bond with a YTM quoted on a semiannual bond basis, this is the form:

$$\text{ModDur}_{\text{SEMI}} = \text{MacDur}_{\text{SEMI}} / (1 + \text{YTM} / 2)$$

EXAMPLE: MODIFIED DURATION

A 5-year, 11% annual coupon bond priced at 86.59 to yield 15% to maturity has a Macaulay duration of 4.03. Calculate the modified duration of this bond.

Answer:

Because it is an annual coupon bond (periodicity = 1), its modified duration can be calculated as follows:

$$\text{ModDur} = 4.03 / 1.15 = 3.50$$

Modified duration provides an estimate for the percentage change in a bond's price given a 1% change in YTM:

$$\text{approximate percentage change in bond price} = -\text{ModDur} \times \Delta\text{YTM}$$

Based on a ModDur of 3.50, in response to an 0.5% increase in YTM the price of the bond should fall by approximately $3.50 \times 0.5\% = 1.75\%$. The resulting price estimate of $86.59 \times (1 - 0.0175) = 85.075$ is close to the value of the bond calculated directly using a YTM of 15.5%, which is 85.092:

$$N = 5; I/Y = 15.5; FV = 100; PMT = 11; CPT \rightarrow PV = -85.092$$

For a semiannual-pay bond, $\text{ModDur}_{\text{SEMI}}$ can be annualized (from semiannual periods to annual periods) by dividing by two, and then used as the approximate change in price for a 1% change in a bond's YTM.

Approximate Modified Duration

We can approximate modified duration directly using bond values for an increase in YTM and for a decrease in YTM of the same size.

The calculation of approximate ModDur is based on a given change in YTM. V_- is the price of the bond if YTM is *decreased* by ΔYTM , and V_+ is the price of the bond if the YTM is *increased* by ΔYTM . Approximate ModDur is given by the following:

$$\text{approximate ModDur} = \frac{V_- - V_+}{2V_0\Delta\text{YTM}}$$

The formula uses the average of the magnitudes of the price increase and the price decrease, which is why $V_- - V_+$ in the numerator is divided by two in the denominator.

V_0 , the current price of the bond, is in the denominator to convert this average price change to a percentage, and the ΔYTM term is in the denominator to scale the duration measure to a 1% change in yield by convention. Note that the ΔYTM term in the denominator must be entered as a decimal (rather than in a whole percentage) to properly scale the duration estimate.

EXAMPLE: CALCULATING APPROXIMATE MODIFIED DURATION

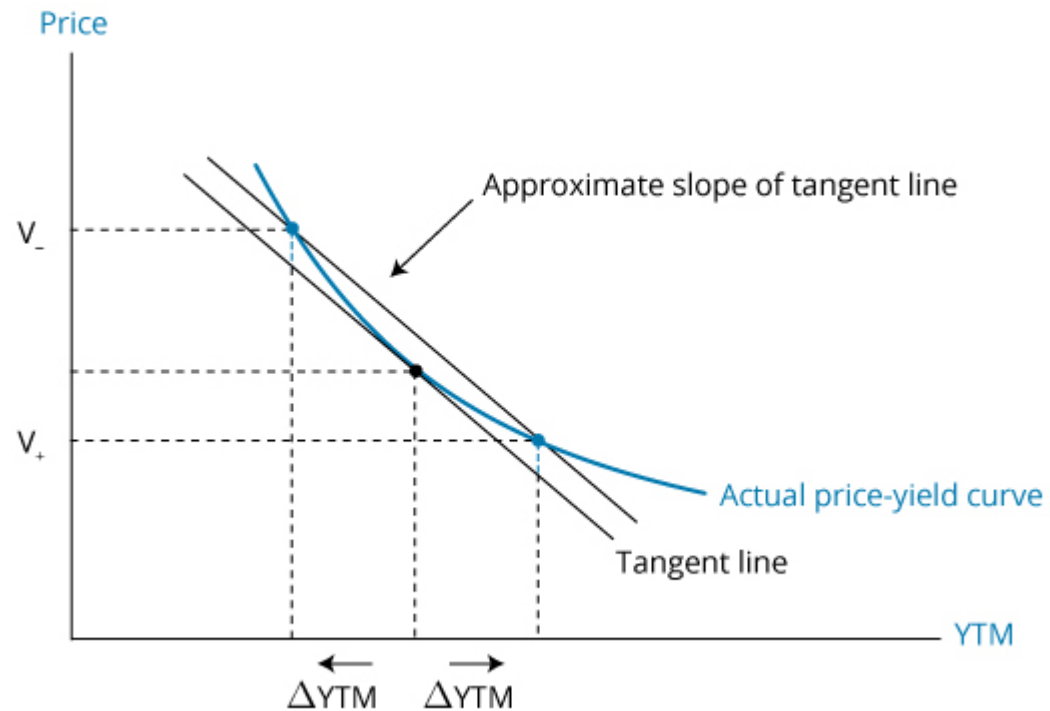
Consider a 5-year, 11% annual coupon bond priced at 86.59 to yield 15% to maturity. If its YTM increases by 50 basis points, its price will decrease to 85.092. If its YTM decreases by 50 basis points, its price will increase to 88.127. Calculate the approximate ModDur.

Answer:

The approximate ModDur is $\frac{88.127 - 85.092}{2 \times 86.59 \times 0.005} = 3.505$ and the approximate change in price for a 1% change in YTM is 3.505%. This result is very close to the ModDur of 3.50 calculated in an earlier example.

Modified duration is a *linear estimate* of the relation between a bond's price and YTM, whereas the actual relation is convex—not a straight line. This means that the modified duration measure provides good estimates of bond prices for small changes in yield, but increasingly poor estimates for larger changes in yield as the effect of the curvature of the price-yield curve is more pronounced. We illustrate this in **Approximate ModDur**.

Approximate ModDur



Money Duration

The **money duration** of a bond position (also called dollar duration) is expressed in currency units:

$$\text{money duration} = \text{annual ModDur} \times \text{full price of bond position}$$

Multiplying the money duration of a bond by a given change in YTM (as a decimal) will provide an estimate for the change in bond value for that change in YTM.

EXAMPLE: MONEY DURATION

1. Calculate the money duration on a coupon date of a \$2 million par value bond that has a ModDur of 7.42 and a full price of 101.32, expressed for the whole bond and per \$100 of face value.
2. What will be the impact on the value of the bond of a 25 basis point increase in its YTM?

Answer:

1. The money duration for the bond is ModDur times the full value of the bond:

$$7.42 \times \$2,000,000 \times 101.32\% = \$15,035,888$$

The money duration per \$100 of par value is:

$$7.42 \times 101.32 = \$751.79$$

$$\text{Or, } \$15,035,888 / (\$2,000,000 / \$100) = \$751.79.$$

2. $\$15,035,888 \times 0.0025 = \$37,589.72$

The bond value decreases by \$37,589.72.

The **price value of a basis point (PVBP)** is the money change in the full price of a bond when its YTM changes by one basis point, or 0.01%. We can calculate the PVBP directly for a bond by calculating the average of the decrease in the full value of a bond when its YTM increases by one basis point, and the increase in the full value of the bond when its YTM decreases by one basis point.

EXAMPLE: CALCULATING THE PVBP

A newly issued, 20-year, 6% annual-pay straight bond is priced at 101.39. Calculate the PVBP for this bond, assuming it has a par value of \$1 million.

Answer:

First, we need to find the YTM of the bond:

$$N = 20; PV = -101.39; PMT = 6; FV = 100; CPT \rightarrow I/Y = 5.88$$

Now, we need the values for the bond with YTM of 5.89 and 5.87:

$$I/Y = 5.89; CPT \rightarrow PV = -101.273 (V_+)$$

$$I/Y = 5.87; \text{CPT} \rightarrow \text{PV} = -101.507 (\text{V}_\text{L})$$

$$\text{PVBP (per \$100 of par value)} = (101.507 - 101.273) / 2 = 0.117$$

For the \$1 million par value bond, each 1 basis point change in the YTM will change the bond's price by 0.117
 $\times \$1 \text{ million} \times 0.01 = \$1,170$.

LOS 57.b: Explain how a bond's maturity, coupon, and yield level affect its interest rate risk.

Other things equal, *bonds with longer maturity* will (usually) have higher interest rate risk. The present values of payments made further in the future are more sensitive to changes in the discount rate used to calculate present value than are the present values of payments made sooner.

We must say *usually* because there are instances where an increase in a discount coupon bond's maturity will decrease its Macaulay duration. For a discount bond, duration first increases with longer maturity and then decreases over a range of relatively long maturities until it approaches the duration of a perpetuity, which is $(1 + \text{YTM}) / \text{YTM}$.

Between coupon dates, if the YTM of a coupon bond remains constant, its Macaulay duration decreases smoothly with the passage of time, and then goes back up slightly at each coupon payment date as the time to the next coupon resets to a full coupon period.

Other things equal, a *higher coupon rate* on a bond will decrease its interest rate risk. For a given maturity and YTM, the duration of a zero-coupon bond will be greater than that of a coupon bond. Increasing the coupon rate means more of a bond's value will be from payments received sooner, so that the value of the bond will be less sensitive to changes in yield.

For floating-rate notes (FRNs) where coupons are periodically reset to a market reference rate (MRR), when interest rates rise, the coupon will also rise, limiting the price risk of the bond. Macaulay duration for an FRN can be calculated as the time to the next coupon reset date at the end of the current coupon period.

Other things equal, an *increase (decrease) in a bond's YTM* will decrease (increase) its interest rate risk. To understand this, we can look to the convexity of the price-yield curve and use its slope as our proxy for interest rate risk. At lower yields, the price-yield curve has a steeper slope, indicating that price is more sensitive to a given change in yield.

Reading 57: Key Concepts

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LOS 57.a

Modified duration is a linear estimate of the percentage change in a bond's price that would result from a 1% change in its YTM:

$$\text{ModDur} = \text{MacDur} / (1 + \text{periodic return of bond})$$

For an expected change in yield, ΔYTM , the expected change in the bond's price is given by the following:

$$\text{approximate percentage change in bond price} = -\text{ModDur} \times \Delta\text{YTM}$$

Modified duration can be approximated by repricing the bond at different yields:

$$\text{approximate modified duration} = \frac{V_- - V_+}{2V_0\Delta\text{YTM}}$$

Money duration is stated in currency units and is sometimes expressed per 100 of bond value:

$$\text{money duration} = \text{annual ModDur} \times \text{full price of bond position}$$

The price value of a basis point is the change in the value of a bond, expressed in currency units, for a change in YTM of one basis point:

$$\text{PVBP} = [(V_- - V_+) / 2] \times \text{par value} \times 0.01$$

PVBP can also be calculated as money duration $\times$ 0.0001.

LOS 57.b

Holding other factors constant, the following are true:

- Duration increases when maturity increases.
- Duration decreases when the coupon rate increases.
- Duration decreases when YTM increases.
- Duration decreases as time passes, but increases slightly on coupon dates.